

BLOEMFONTEIN CAMPUS

CSIS3754

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

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SEMESTER TEST 2

22 May 2024

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MODERATOR(S): 1. Prof. T. du Toit

TIME: 180 min

MARKS: 90

INSTRUCTIONS:

- This question paper has three questions:
 - Answer Question 1 on the answer sheet provided. (Closed Book)
 - Create and submit a Jupyter Notebook for Question 2. (Open Book)
 - Create and submit a Jupyter Notebook for Question 3. (Open Book)
- Submit the two notebooks to the following Blackboard links, respectively:
 - CSIS3754 -> Assessments -> Semester Test 2 -> Question 2.
 - CSIS3754 -> Assessments -> Semester Test 2 -> Question 3.

Question 1 [20 Marks - 40 Minutes] (Closed Book)

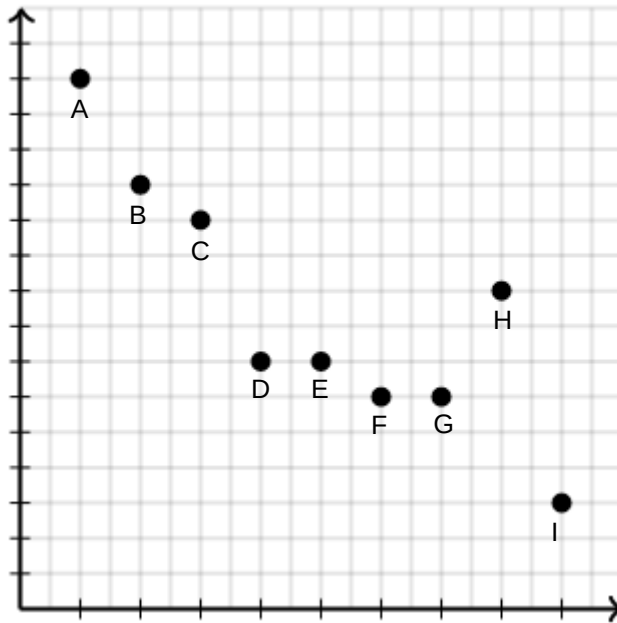
1.1 Consider the following learning curves of two machine learning models. On the left, a learning curve of a Support Vector Machine classifier and on the right, a learning curve of a K-Nearest Neighbour classifier:



1.1.1 Which model would most probably provide better prediction results? (1)

1.1.2 Motivate your answer in 1.1.1 by explaining why the relevant model would provide better prediction results. (3)

1.2 Consider the following scatter plot:



1.2.1 Draw a dendrogram to illustrate how the data points can be grouped by means of Agglomerative Hierarchical clustering by using the minimum similarity algorithm (single-linkage algorithm). (4)

1.3 Consider the data table below that can be used to train a decision tree capable of determining whether a person is likely to suffer a heart attack depending on the person's blood pressure and age:

High blood pressure (HBP)	Younger than 48 (Age < 48)	Heart attack (HA)
Yes	Yes	Yes
No	Yes	No
Yes	Yes	No
No	Yes	No
Yes	No	Yes
Yes	Yes	No
Yes	No	Yes

1.3.1 Using the Gini Impurity metric, determine the root feature of the decision tree. Show all your calculations. (12)

[20]

Question 2 [50 Marks - 100 Minutes] (Open Book)

Please log into Blackboard and go to CSIS3754 -> Assessments -> Semester Test 2 -> Question 2 and download the data file named *fake_currency_data.csv*. The data contains information about counterfeit and genuine printed currency (money). Create a Jupyter notebook that contains the Python code to construct machine learning classifiers that are able to determine whether printed currency is genuine or

counterfeit, depending on the feature values for that specific printed currency. Name the notebook “CSIS3754_YourStudentNumber_ST2-2_2024”.

The Jupyter notebook should have the following functionality (NB!!! - Show the changes in the data after each operation):

- 2.1 Import the data file into a dataframe named “currency” with the column headings as contained in the data file. (1)
- 2.2 The values in the *Country* column refers to the country in which the currency was issued. Therefore, change the column heading “Country” to “CountryOfOrigin” to provide a better description for the values contained in this column. (2)
- 2.3 Inspect the data by: (3)
 - Generating a concise summary of the dataframe indicating the index, data type, columns, non-null values and memory usage.
 - Generating a statistical summary of all the features.
 - Displaying the number of unique values for each feature.
- 2.3.1 Discuss the inferences that can be drawn from the concise summary, the number of unique values, and the statistical summary of the dataframe, as well as possible issues that can be identified from the data inspection? (Print your answer in the Jupyter notebook) (5)
- 2.4 Visualise the data by using the seaborn library to create the following plots: (2)
 - Create boxplots for the following numeric features:
 - Weight
 - Length
 - Width
 - Thickness
- 2.4.1 Discuss the issue that can be deduced from the relevant plots: (Print your answer in the Jupyter notebook) (1)
- 2.4.2 Fix the issue identified in 2.4.1 by: (3)
 - Performing the necessary operations to fix the issue.
 - Motivating your chosen method for fixing the issue.
 - Confirming that the issue was fixed successfully by recreating the boxplots for the specified numeric features.
- 2.5 Due to the number of samples that are currently contained in the dataframe it will take a long time to train several machine learning models. To avoid this, the sample size of the dataframe needs to be reduced. Therefore, select a random number of samples (without replacement) from the *currency* dataframe to be used from this point on. (2)
- 2.6 Determine whether the data is balanced with regard to the number of counterfeit and genuine currency by: (3)
 - Creating a pie chart that shows the percentage of records for counterfeit vs. genuine currency.
 - Percentages displayed on the pie chart should be rounded to 1 decimal.
- 2.7 Discuss whether the data is currently balanced and if further balancing is necessary. (1)

- 2.8 Convert the text values in the dataframe to numeric values: (6)
- Convert the columns that contain only two unique text values to binary values [0, 1].
 - Convert the columns that contain more than two unique text values to a vector space using one-hot encoding.
- 2.9 Confirm that there are no more columns of type object. (1)
- 2.10 Define X and y and create a training / testing dataset of 80% / 20%: (6)
- X = Data features.
 - y = Counterfeit.
 - Determine the dimensions of X_train and X_test.
 - Ensure that all pre-processing is complete.
- 2.11 Train the following classifiers and determine the best model by using k-fold cross-validation: (8)
- Naïve Bayes (default values).
 - Logistic Regression (default values).
 - Support Vector Machine (gamma = auto).
 - Set k = 5 for the k-fold cross-validation.
 - Report the cross-validated training accuracy and F1 scores for all the classifiers.
 - Draw a learning curve for each of the classifiers.
- 2.12 Select the model that produced the highest F1 score to do the following: (3)
- Make predictions using the test dataset.
 - Provide the test accuracy score, confusion matrix and classification report of the model.
- 2.13 Discuss the selected model's metrics and scores with regard to the following (Print your answer in the Jupyter notebook): (3)
- Discuss the model's training accuracy in comparison to its testing accuracy.
 - Discuss the meaning of the precision, recall and F1-scores.
- [50]

Question 3 [20 Marks - 40 Minutes] (Open Book)

Please log into Blackboard and go to CSIS3754 -> Assessments -> Semester Test 2 -> Question 3 and download the data file named *gin.csv*. The data contains information about different types of gin with regard to their chemical composition.

Create a Jupyter notebook that contains the Python code to construct an unsupervised machine learning model that will be able to split different types of gin into different groups according to their chemical composition. Name the notebook "*CSIS3754_YourStudentNumber_ST2-3_2024*".

The Jupyter notebook should have the following functionality (NB!!! - Show the changes in the data after each operation.):

- 3.1 Import the data file into a dataframe named "gin". (1)
- 3.2 Make sure there is no zero values in the dataset. (1)

- 3.3 Use k-Means clustering to train a machine learning model that will have the capability to distinguish between different types of gin. (11)
- Use only the following features:
 - Alcohol
 - Malic_Acid
 - Ash_Alcanity
 - Proanthocyanins
 - Hue
 - Make sure all pre-processing is complete.
 - Determine the best value for k by calculating and plotting the silhouette score.
 - Train the model, predict the labels and print the labels of the trained model.
 - Print the cluster centres that were identified by the trained model.
- 3.4 Visualise the data using the seaborn library: (5)
- Create scatter plots between the different pairs of features. This should give you 10 separate scatter plots.
 - Show the clusters that were identified by the model within the data by means of different colours.
 - Also plot the centroids of the clusters on the same scatter plot.
- 3.5 Evaluate and discuss the optimal k-value that was obtained from the silhouette score in relation to the number of clusters that are visible within the scatter plots. (Print your answer in the Jupyter notebook) (2)
- [20]

End of Paper